Evaluating the Impact of Augmenting the VAEP Model with 360-Degree Freeze-Frame Spatial Context

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Abstract

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1. Introduction

The adoption of analytics in association football has grown significantly in the last couple of decades (Goes et al., 2021; Olthof & Davis, 2025; Schaerlaeckens, 2025; Wakelam et al., 2022), enabling football clubs to leverage insights to inform tactical decisions, coaching and recruitment. Much of this innovation has involved the development of metrics used to determine the value provided by a particular action made by a player in terms of goal-scoring probabilities. One such metric is Valuing Actions by Estimating Probabilities (VAEP) (Decroos et al., 2019).

VAEP considers the probability of scoring or conceding within the next k events to assign a value between -2 and 2 to each action. VAEP, like other such metrics, typically uses action-centric features such as the action's start and end location and type (carry, pass, shot) in estimating scoring/conceding probabilties, but this approach does not explicitly model the wider spatial context (Sá-Freire, 2025). The spatial context - the position of teammates and opponents around the actor - of an action determines the options available to a player, showing viable passing options or space to carry into, as well as the proximity of opposition defenders applying pressure or blocking options.

The StatsBomb Open Data repository (StatsBomb, 2026a) provides events (e.g. carries, passes, shots) data for matches, along with *three-sixty* freeze-frames which indicate the positions of players within proximity of the acting player. Such data can be used to incorporate spatial context as a predictive feature. To illustrate this, **Figure 1** compares a pass event from a match with and without the surrounding spatial context. Without the spatial context, this pass appears to offer limited progression towards the opponent's goal. Traditional VAEP would consider how passes of this trajectory tend to result in scoring/conceding, but the addition of spatial context differentiates this particular pass from others of similar trajectory. **Figure 1** shows how the left hand side of the actor is blocked by multiple opponents (red) and this pass moves the ball into space on the right flank to open an attacking opportunity (in this case, a long-ranged shot)[1]. Spatial context could allow finer-grained differentiation, rather than aggregating all passes of this trajectory.

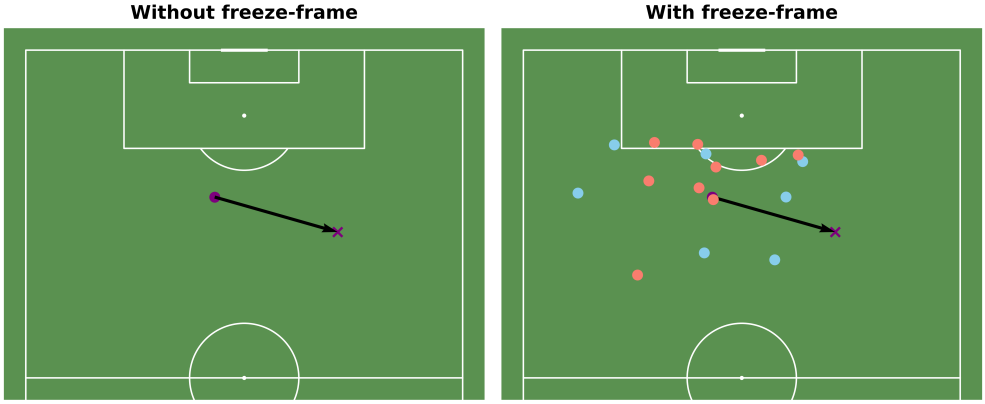


Figure 1: Pass Event With and Without Freeze-Frame

[1]The pass event id is '23b6b26a-293c-4d5e-9889-5a5ce0b7ad8c' and the subsequent long-ranged shot event id is 'd9d27d20-35bc-42bc-ac53-059ad519f781'

2. Research Objectives

The general aim of this project is to determine to what extent the addition of spatial context features enhances and alters the standard VAEP model. This entails the following objectives:

- Develop representations of spatial context from StatsBomb 360 freeze-frame data using feature engineering and deep learning techniques.
- Train and evaluate VAEP models, both with and without the inclusion of spatial context representations, using supervised classification algorithms.
- Compare and interpret these VAEP models using appropriate peformance metrics and an exploration of feature importance.

3. Background

In soccer analytics, *actions* refer to a *type of event* within a match which is caused by a player (e.g. a pass), with *events* being the broader set of significant occurrences in a match (e.g. passes, but also substitutions, fouls etc.) (Decroos, Bransen, Van Haaren, & Davis, 2019). Typical performance metrics focus on action outcomes, such as goals, assists and pass completion rates. Whilst such metrics can be used to determine the value of some actions and are useful for summarising player ability, soccer analytics has developed metrics to help generalise this notion of action value. Moreover, models used to assign value to actions have been enhanced by incorporating the locations of teammates and opponents, leading to improved performance and interpretability.

As such, this section will introduce the VAEP metric and outline how spatial context has been successfully incorporated into other action valuation metrics, motivating this project's research objectives.

3.1. Valuing Actions by Estimating Probabilities (VAEP)

Decroos, Bransen, Van Haaren, & Davis (2019) introduce Valuing Actions by Estimating Probabilities (VAEP). VAEP values actions in terms of the probability of scoring and the probability of conceding within the next k actions. Equation 1 defines the VAEP of an action a_i for team x.

$$V(a_i, x) = \Delta P_{scores}(a_i, x) - \Delta P_{concedes}(a_i, x), \quad V(a_i, x) \in [-2, 2] \tag{1}$$

If we consider a match as a sequence of m on-ball actions $[a_1, ..., a_m]$, we can view some game state S_i which corresponds to some subset of actions $[a_1, ..., a_i]$ and determine how the last action in the game state a_i changed the probability of team x scoring and conceding within the next k actions. The change in the probability of scoring (and conceding) from the previous game state $S_{i-1} = [a_1, ..., a_{i-1}]$ to the current game state S_i can be determined by simply finding their difference:

$$\begin{aligned} \Delta P_{\text{scores}}(a_i, x) &= P_{\text{scores}}(S_i, x) - P_{\text{scores}}(S_{i-1}, x) \\ \Delta P_{\text{concede}}(a_i, x) &= P_{\text{concede}}(S_i, x) - P_{\text{concede}}(S_{i-1}, x) \end{aligned} \tag{2}$$

These two probabilities are estimated using two supervised classification algorithms, one which estimates scoring, and the other conceding, within the next k actions. The feature sets used across these two models are identical, all that varies is the labels (scoring or conceding). Decroos, Bransen, Van Haaren, & Davis (2019) used a feature set which corresponds to the

Soccer Player Action Description Language (SPADL) format involving 9 features, most notably the action's start and end locations, and the type of action. The authors also add "complex features", such as the time elapsed between consecutive actions, as well as "game context features", such as the scoreline at the time of the action and the amount of time remaining.

Decroos, Bransen, Van Haaren, & Davis (2019) initially compared logistic regression, a random forest, XGBoost and CatBoost, finding CatBoost to perform best across both the scoring and conceding models in terms of the Brier Score and the Area Under the Receiver Operating Characteristic Curve (AUC-ROC). Decroos, Bransen, Van Haaren, & Davis (2019) also extend the VAEP metric by using it to provide player ratings, where the VAEP of each action in a player's set of actions A_p^T across some time frame T are considered cumulatively, and normalised per-90 minutes played during T.

$$\text{rating } (p) = \frac{90}{m} \sum_{a_i \in A_p^T} V(a_i) \tag{3}$$

The authors compare these VAEP player ratings against traditional metrics like goals and assists per-90, as well as market valuations of players. The authors found the top-10 highest players ranked by VAEP captured players typically considered to be amongst the best in the Premier League, despite them not appearing in the top-10 goals + assists per-90 ranking. The top-10 VAEP ranked list also had a higher overall market valuation than the top-10 goals + assists per-90 ranked list (€$1,110$ Million vs $947 Million$), which suggests VAEP more comprehensively represents player value compared to traditional metrics.

Decroos & Davis (2020) extend their previous work by improving model interpretability. Given that the initial findings from Decroos, Bransen, Van Haaren, & Davis (2019) suggest the use of gradient boosting algorithms for VAEP model classifiers, these models are more difficult to interpret than methods such as logistic regression. As such, Decroos & Davis (2020) built a generalised additive model (GAM) on the top-10 most important features from the original VAEP feature set, as determined by the average *gain* (for the present classifiers, this is the reduction of Gini impurity - see Section 4.5.5) that feature imports on the model across all splits involving that feature. The authors found a very small drop in classifier performance: a 0.2% increase in the normalised Brier score compared to XGBoost on the original VAEP feature set, and a 0.1 increase on the top-10 most important feature set. The authors argue the large gains in utility came from the added interpretability from being able to assess each univariate function which comprised their GAM model. This enables a direct observation of how each function contributes to its prediction across the domain of that function. For example, *distance to opponent goal* constituted one of the top-10 features, and figure 2 in Decroos & Davis (2020) shows how the goal scoring probability changes across the domain of this feature's function. Matching intuition, a higher goal scoring probability is observed for lower distances to the opponent's goal, with this probability dropping exponentially as distance increases. This added interpretability helps to demonstrate the greater tactical understanding which soccer analysts may benefit from when using a GAM-based VAEP model, and that such interpretability may be gained without a significant loss in performance.

One potential limitation of the VAEP model is that it lacks a representation of wider spatial context from the positions of teammates and opponents during actions (Sá-Freire, 2025), due to its use of event data only. Every action taken by a player exists within a spatial context which determines the options available to a player, influencing their decision making. The use of spatial context in VAEP models is also understudied, despite its addition having been successfully applied to a number of alternative action value metrics.

3.2. Applications of Spatial Context to Action Value Metrics

3.2.1. *Expected Goals (xG)*

Expected goals (xG) is an approach commonly used for assigning value to individual shots (Green, 2012; Macdonald, 2012). General to xG approaches is the representation of the value of some shot as the probability that it results in a goal. These goal probabilities are estimated via various classification algorithms such as logistic regression, gradient boosting and deep learning methods (Mead et al., 2023). Features used in xG models typically include *action-intrinsic* variables such as distance/angle to the centre of the opponent's goal and the body part used, along with *match-context* variables such as the current match scenario (e.g. set piece, open play) (Anzer & Bauer, 2021; Mead, O'Hare, & McMenemy, 2023; Robberechts & Davis, 2020).

Cefis & Carpita (2025) incorporate spatial context into the xG model by considering the number of players ahead of the shooter (x co-ordinate greater than the shooter's) within a fixed radius (y co-ordinate $\pm 10\%$ of the shooter's). From this count, they derive two binary variables `D1.OpponentsPlayer` and `D2.OpponentsPlayer`, where the former equals 1 when the number of players in this area is greater than 0, and the latter equals 1 when the number of players in the area is greater than 1. They also consider the x and y co-ordinates of the opponent's goalkeeper (`GK x coordinate`, `GK y coordinate`). Table 3 in Cefis & Carpita (2025) show the coefficients and odds ratios of each feature in their logistic regression xG model across their 4 different techniques for addressing class imbalance. `D1.OpponentsPlayer`, `D2.OpponentsPlayer`, and `GK x coordinate` all have negative coefficients, indicating a reduction in the log odds of the shot resulting in a goal. This is particularly pronounced for `D2.OpponentsPlayer`, with a coefficient range of -6.13 to -5.69, indicating that the probability of scoring (i.e. xG) is reduced by around 99.73% (because $1 - e^{(-5.69 + -6.13)/2} \approx 0.9973$) when there is more than 1 opposition defender within proximity of the shooter. This finding matches intuitions about the influence of the opposition's positioning on goal probability and highlights how the addition of spatial context during modelling can enhance the xG metric.

3.2.2. *Expected Threat (xT)*

A further action value metric is expected threat (xT). xT measures the value of actions based on where on the pitch the action moves the ball from and to. Singh (2018) derived the following equation for calculating the xT of a given location on the pitch:

$$\mathrm{xT}_{x,y} = \left(s_{x,y} \times g_{x,y}\right)\left(m_{x,y} \times \sum_{z=1}^{Z}\sum_{w=1}^{W} T_{(x,y)\to(z,w)}\mathrm{xT}_{z,w}\right) \tag{4}$$

If we discretise the football pitch into some $Z \times W$ (Singh (2018) uses $Z = 16$ and $W = 12$) set of equally-sized square zones (x, y), we can consider three possible attacking options when the ball is in some zone (x, y): shoot, carry or pass. $\left(s_{x,y} \times g_{x,y}\right)$ quantifies the value of shooting from zone (x, y) by weighting the probability of scoring $g_{x,y}$ by the probability of shooting $s_{x,y}$. The other two options, carrying or passing, involve moving the ball from (x, y) to (z, w). The transition matrix $T_{(x,y)}$ provides the probablity of moving the ball from zone (x, y) to any zone (z, w) after 1 time step. If we consider all $Z \times W$ possible options and multiply their transition probabilities by the value of its respective (z, w) zone, we can take their sum and weight it by the probability of actually moving the ball to another zone when in (x, y). This provides a quantification of passing or carrying the ball, which when multiplied by the shot-probability weighted xG of zone (x, y) provides the xT of the ball being at zone (x, y). However, given that Equation 4 assumes that the value of each zone $\left(\mathrm{xT}_{z,w}\right)$ is already known a-priori, we must

estimate its value. The xT model thus begins by setting every zone (x, y) to have an xT of 0. This simplies Equation 4 to essentialy be the xG of zone (x, y), weighted by how likely players are to actually shoot from that zone:

$$\mathrm{xT}^{(0)}_{x,y} = \left(s_{x,y} \times g_{x,y}\right) \tag{5}$$

We can then take the xT values from this first iteration and recalculate them iteratively until convergence. Singh (2018) suggests 4–5 iterations are sufficient but that this may vary across datasets. To then assign xT to an action, we simply subtract the xT of the end location of the action (i.e. the xT of zone (z, w)) by the xT of the start location (the xT of zone (x, y)). This approach is useful for allowing us to determine the value of any on-ball attacking action, not just shots. We can also quantify genuinely valuable actions at any point during a chain of actions (e.g. an incisive pass during a sequence which ends in a goal), rather than suffering from a bias towards teminating actions (goals, assists).

Everett et al. (2022) introduce spatial context to the xT transition matrix $T_{(x,y)}$ via a pitch control (Spearman et al., 2017) and interception model (Burriel & Buldú, 2021) which was used to train a convolutional neural network (CNN) to produce the transition probabilities for their contextual transition matrix $T^{\star}_{(x,y)}$. The pitch control model considers the probability of a player controlling the ball as a function of their distance from opposition players, whilst the interception model introduces an additional transition probability, that being the probability of the ball being intercepted when moving from zone (x, y) to its intended destination (z, w). Everett, Beal, Matthews, Norman, & Ramchurn (2022) found the introduction of spatial context yielded a modest improvement in terms of model fit, reducing the log likelihood of the xT model from -5.137 to -4.151. Greater gains were found in its useful applications to tactical analysis, such as enabling the authors to apply a Basin-Hopping optimiser (Wales & Doye, 1997) to find more optimal defensive shapes than which actually occurred during actions, using their derived Threat Above Expected (TAx) metric as the algorithm's loss function.

3.2.3. *Expected Possession Value (EPV)*

Expected possession value (EPV) (Fernandez et al., 2021) is an action valuation metric, derived from other sports such as basketball (Cervone et al., 2016), which exploits tracking data to assign value to a game state (e.g. during an action) at time t via a consideration of all possible subsequent paths it can take.

$$\mathrm{EPV}_t = \sum_{a \in A} \mathbb{E}[G \mid A = a, T_t] \quad \mathbb{P}(A = a \mid T_t) \tag{6}$$

Equation 6 shows that the EPV at time t for some action is determined by the expected value of G (typically a goal being scored or not) given some action a has occurred in some spatiotemporal context T_t, multiplied by the probability of that action a occurring in the spatiotemporal context T_t. For each action a considered in the action set A, we sum these expected values to provide an EPV of some game state at time t. Thus EPV can be applied to any game state, not simply at the time at which discrete match events or actions occur.

We can see from the term T_t that EPV relies upon spatial context to operate. Fernandez, Bornn, & Cervone (2021) list a number of spatial features used in the EPV model, such as pitch control, number of players within 3m of the ball location, and the number of players within the area circumscribed by the ball and the two posts of the opponent's goal. EPV is thus inherently sensitive to spatial context in its valuation of actions. The utility provided by EPV's

incorporation of spatial context is explored by Fernandez, Bornn, & Cervone (2021). For example, they use EPV to determine the on-ball (passes with positive EPV) and off-ball (pitch locations with positive EPV if a pass were to arrive there) advantage Liverpool in 2014/15 had during possession when up against different opponent formations. This analysis shows both where Liverpool were successful and unsuccessful in their passing decision (on-ball) and also where valuable regions of the pitch are to direct passes towards during build-up (off-ball). Such an analysis is extremely useful for tactical decision making as, in response to opposition pressing structures, teams can evaluate where on the pitch they should look to progress the ball towards to be most effective during build-up play, and to evaluate how successful their current approach is.

3.2.4. *Summary*

To summarise, the discussed applications demonstrate the utility provided by incorporating spatial context into action valuation models. Given the lack of investigation into augmenting VAEP models with wider spatial context, this project addresses this research gap and explores what role representations of spatial context play in altering model behaviour and, potentially, performance.

3.3. Representing Spatial Context

3.3.1. *Manual Spatial Feature Engineering*

Previous research has represented spatial context using features engineered from spatial data.

As described in Section 3.2.1, Cefis & Carpita (2025) derived two binary features `D1.OpponentsPlayer` and `D2.OpponentsPlayer` from player x and y co-coordinates, indicating the number of players within the immediate frontal proximity of the shooter.

Peters et al. (2024) developed an Expected Pass Turnovers (xPT) metric, used to measure the probability that a given pass would result in a turnover (conceding possession to the opposition) which can be used to assess how player's passing ability during some time interval compares to what would be expected on average, given the set of passes made by that player and the contextual factors relating to those passes. The authors derive xPT values from a binary logistic mixed-effects model trained on a feature set which included two features derived from StatsBomb *three-sixty* freeze-frames: "three pressure radii" (the number of opponents within three radii of varying, but unspecified length around the ball carrier) and number of "unmarked teammates" (teammates with no opponents within 2m of them). The authors found the addition of these "positional" features improved model AUC by 6.1 points

Amichay et al. (2025) measured the areas of convex hulls and inner layers to represent team shapes. A convex hull, in this context, is the area inscribed between the outer-most positioned players on the pitch for a given team. A larger area corresponds to a wider/longer team shape, whereas a smaller area is narrower/shorter. Since this convex hull only considers the outer-most players, a second convex layer can be created from the players who do not contribute to the hull. The ratio of these two areas forms the author's derived Layer Ratio (LR) metric, which summarises the compactness of the team shape.

3.3.2. *Spatial Embeddings: Set Transformer*

An alternative way to represent spatial context is in the form of spatial embeddings. A spatial embedding is a real-valued vector used to represent a set of geometric objects (in this case, player co-ordinates and their characteristics) in a fixed d-dimensional space. These embeddings

can represent non-linear relationships between objects and thus may serve to represent patterns, like defensive and offensive team structures during actions.

However, as seen in Figure 2, StatsBomb *three-sixty* data does not identify individual players within each freeze frame (except from the actor). Thus, to create embeddings for player positions, we must use a neural network which has an input layer which is *permutation invariant*. That is, the output of the network (i.e. the embeddings) should not depend on the order in which features appear in the input layer. This would enable the locations of players in each freeze-frame to be inputted into the network in any order and still produce the same embedding. It would also be useful for such a network to attend to spatial relationships that may exist across freeze-frames, such as defensive shapes.

One useful technique for developing a permutation-invariant network for spatial embeddings is a *set transformer* (Lee et al., 2019). A set transformer is a variation on the standard transformer architecture from Vaswani et al. (2017) which can be used to process sets such that the output of the transformer is invariant to how elements in the set are permuted when entering into the model.

Transformers use multihead self-attention layers in a traditional encoder-decoder architecture. Multihead self-attention works by projecting an input sequence $X \in \mathbf{R}^{n \times d}$ into different learned linear projections of the input embeddings: $Q = XW_Q, K = XW_K, V = XW_V \in \mathbb{R}^{n \times d}$. Each row corresponds to a token representation in a different learned subspace. The self-attention mechanism determines how much each token should attend to other tokens, including itself, by computing similarity between so-called *query* vectors and *key* vectors. If the similarity between a *query* and a *key* is high, the corresponding *value* vector contributes more strongly to the output representation. This is computed for all tokens simultaneously, producing pairwise interactions between all elements in the sequence, including self-interactions. The attention mechanism achieves this via scaled dot-product attention:

$$\text{Attention } (Q, K, V) = \text{softmax } \left(\frac{QK^\top}{\sqrt{d_k}} \right) V \tag{7}$$

Multihead attention then involves performing this scaled dot-product attention in parallel h times, each with their own set of Q, K, V matrices. Outputs are then concatenated and fed into a standard feedforward neural network.

The multi-head attention mechanism of standard transformers is adapted in set transformers via *induced set attention blocks* (ISABs). An ISAB performs self-attention between set elements by introducing *inducing points* which contain parameters learned during training. These inducing points are some fixed-sized vector I and attend to the inputted set X to produce a transformed set H. The input set X then attends to H to produce a final set of n elements. This method improves computational efficiency as the size of the inducing points vector $|I|$ can be set to be smaller than the input size $|X|$, thus avoiding standard pairwise self-attention and reducing time complexity from $O(n^2)$ to $O(nm)$ where $m < n$.

Important for the purposes of this present project, set transformers also enable permutation invariance by dropping position encodings of input set elements and pool outputs via pooling with multihead attention (PMA). PMA works similarly to multihead attention but instead pools the representations of each element of the set into k vectors, each providing a representation of the whole set. k is typically set to 1, provding a single representation of the set, but $k > 1$ is permitted. The queries Q are a fixed set of learned *seed vectors* S. Reordering the set will thus result in the same pooled representation via PMA, enabling permutation invariance.

Applications of set transformers in soccer analytics have typically fallen in the domain of trajectory prediction. Kim et al. (2023) apply set transformers in the context of soccer analytics. Kim, Choi, Kim, Yoon, & Ko (2023) use set transformers to encode representations of player trajectories from tracking data snapshots to infer ball trajectories. Capellera et al. (2024) similarly developed *FootBots* using set transformer architecture to improve player and ball motion prediction over traditional models. Whilst this domain deviates from this project, these studies do highlight the appropriateness of using set transformers to encode player positions.

This project seeks to use set transformers to produced fixed-sizes spatial embeddings for each team's set of players within each freeze-frame sample. This can be achieved by first converting each individual co-ordinate pair, along with its characteristics (teammate, goalkeeper, actor), into an embedding via an initial input projection sublayer. Each set of individual co-ordinate embeddings for each team in each freeze-frame can then be passed as an input tensor to the set transformer which will produce embedding representations of each set of co-ordinates. The initial embedding of co-ordinate pairs is necessary to ensure the set transformer has enough dimensions to learn from whereas simply having 2 dimensions (i.e. x and y) would be insufficient to capture multiple non-linear relationships.

4. Methodology

4.1. Datasets Outline

The project uses data from StatsBomb's Open Data repository. This repository has "made certain leagues of StatsBomb Data freely available for public use for research projects and genuine interest in football analytics" (StatsBomb, 2026a). The StatsBomb Open Data repository provides the following datasets, and data specification documents exist which outline variables, data types and variable descriptions for each dataset:

- **competitions.json:** A single JSON file for competition summary information.
- **events:** A directory containing a single JSON file for each match, detailing each event which takes place within that match (e.g. each pass, shot, carry etc.).
- **lineups:** A directory containing a single JSON file for each match, providing information about each player who participated in that match (e.g. their position, shirt number, nationality).
- **matches:** A directory containing a single JSON file for a season in a given competition, providing summary information about each match played in that season (e.g. match date, home/away teams and scores).
- **three-sixty:** A directory containing a single JSON file for each match, providing data for each event in a given match detailing the position of each player in the broadcaster's frame at the moment of that event.

Decroos, Bransen, Van Haaren, & Davis (2019), along with VAEP, introduce the Soccer Player Action Description Language (SPADL) as a data engineering format to enable "human-interpretable, simple and complete" records of actions within football matches. A variant of this format is used in this project for the baseline feature set, that being Atomic-SPADL (Davis et al., 2020). Atomic-SPADL represents an action without considering its result (success vs failure), unlike SPADL which includes the result. An example of the effect this has on action representation is that a pass (one action under SPADL) is separated into two actions: the giving of the pass, and the reception of the pass. An additional "interception" action type is introduced to enable incomplete passes to be captured. Atomic-SPADL therefore introduces more precision in how actions are represented as actions are now indivisible (hence atomic). This atomic-SPADL format is then augmented in this project by the addition of engineered spatial features.

The project uses *events* and *three-sixty* data corresponding to 292 matches from UEFA Men's Euro 2020 & 2024, UEFA Women's Euro 2022 & 2025, FIFA Men's World Cup 2022, FIFA Women's World Cup 2023. Unlike most competitions in this repository, *three-sixty* data is available for every match in this sample, enabling player positional context to be incorporated into feature engineering for 1,052,116 match events.

4.1.1. *Limitations of Datasets*

Football analysts often makes use of tracking data (Goes, Meerhoff, Bueno, Rodrigues, Moura, Brink, Elferink-Gemser, Knobbe, Cunha, Torres, & Lemmink, 2021). Rather than considering discrete time events, tracking data provides a continuous record of player and ball position for the duration of a match. This enables analysts to model actions in matches as continuous processes, accounting for the position and movements of the entire team, unlike with events data. Such tracking data is often proprietory and is thus not available for public use (Bassek et al., 2025). This limits the scope of the present project to viewing events as discrete in time, thus losing some surrounding context which may be informative (e.g. player movement direction and velocity).

Events data is nonetheless relatively comprehensive, covering a much larger sample of matches than the limited publicly-available tracking data (Olthof & Davis, 2025). Furthermore, the use of StatsBomb's *three-sixty* data does address this limitation somewhat by providing snapshots of proximal player positions at the point of each event, allowing some inference to be made regarding team positioning during match events. However, these snapshots do not necessarily capture every player's location during an event, just those within the view of the broadcast camera (StatsBomb, 2026a).

A further limitation is with the use of SPADL formatted data. Recent work by Jo et al. (2026) has shown that some events data contains logical inconsistencies (i.e. sequences of events which are not logically possible), which the SPADL format does not correct for. The authors develop the Verified Event Data Format for Reliable Soccer Analytics (VERSA) to address this issue using a state-transition model to automatically identify and correct logical inconsitencies, improving performance on VAEP model training. However, their results on StatsBomb data suggest that the atomic-SPADL format still slightly outperforms VERSA. This result, along with the increased precision in action representation mentioned earlier, motivates the use of the atomic-SPADL format in this project, but it is recognised that the potential data integrity issues highlighted by Jo, Kang, Lee, Lee, Bauer, & Ko (2026) will not be explicitly handeled as implementing VERSA is non-trivial and thus beyond the scope of this project.

4.2. Data Collection

Data was obtained from the StatsBomb Open Data Repository via the `statsbombpy` (StatsBomb, 2026b), `socceraction` (Socceraction Developers, n.d., Decroos, Bransen, Van Haaren, & Davis (2019)), and requests (, n.d.) libraries. The `statsbombpy` and `socceractions` libraries were used to collect raw events data from the matches in each competition, whilst `requests` was used to collect *three-sixty* freeze-frames from the StatsBomb Open Data Repository directly. Events data was then merged with freeze-frame data and each competition's data was saved to an individual parquet file.

4.3. Data Preprocessing

With StatsBomb events and *three-sixty* data collected, the first preprocessing step requires converting these events into SPADL-formatted actions. This was achieved by passing each match's set of events to the `convert_to_actions` function from `socceraction`. This also required

the rejoining of freeze frame data, since this preprocessing function only returns SPADL formatted data, not custom features.

To then convert these actions to the atomic-SPADL format, the `convert_to_atomic` function similarly takes each match's set of actions and returns them in atomic-SPADL format. The atomic actions of each match from each competition are then saved to parquet files, one for each competition.

With the derived atomic-SPADL actions, spatial context features are subsequently added during data engineering.

4.4. Data Engineering

4.4.1. *Manually-Engineered Spatial Context Features*

```
{
    "teammate": boolean,
    "actor": boolean,
    "keeper": boolean,
    "location": [float, float]
}
```

Figure 2: StatsBomb *three-sixty* freeze-frame JSON schema

Figure 2 outlines the JSON schema for StatsBomb *three-sixty* freeze-frames. Each freeze-frame JSON object has an associated `event_uuid` to enable joins with events data, and a `visible_area` polygon which indicates the viewing area from the broadcast camera at the moment the action took place. We can see from the available data, that it is possible to construct the opposition proximity metric from Cefis & Carpita (2025) and a partial convex hull area or LR as explored by Amichay, Silva, Brito, & Marcelino (2025). Given that the sample in this project applies to on-ball actions, not just shots, the opposition proximity metric from Cefis & Carpita (2025) is adapted to consider a 360° radius around the player, rather than a frontal semi-circle.

Thus, for this project, the following spatial features were engineered for teammates and opponents, respectively, using *three-sixty* data. These are both inspired by the literature and also chosen by myself:

- Euclidean distance to nearest teammate (meters)
- Euclidean distance to nearest opponent (meters)
- Number of teammates within 5 meters of the actor
- Number of opponents within 5 meters of the actor
- Average Euclidean distance to teammates in freeze-frame (meters)
- Standard deviation of Euclidean distance to teammates in freeze-frame (meters)
- Average Euclidean distance to opponents in freeze-frame (meters)
- Standard deviation of Euclidean distance to opponents in freeze-frame (meters)
- Teammate convex hull area (based on players within freeze-frame) (m^2)
- Opponent convex hull area (based on players within freeze-frame) (m^2)

To outline the rationale here, opponent spatial metrics are intended to provide a proxy measure of *pressure* applied to the actor and *defensive shape*, which imposes limitations upon where the ball can be progressed to by introducing a risk of losing possession. Teammate spatial metrics provide a proxy measure of *pass availability* for the actor and *rest defense* - the positional structure of a team whilst in possession to prepare for potential counter-attacks (Coaches' Voice, 2024).

For *number of teammates/opponents within 10 meters of the actor*, whilst a radius of 5 meters somewhat arbitrary, Bischofberger et al. (2026) argue that a "lower radius might risk missing actual pressure and involvement while a higher radius might lead to a weaker attribution between players and stronger intra-team correlations" p.5. Such an approach to approximating opposition pressure or teammate pass availability is also considered relatively simplistic, with more sophisticated approaches such as pitch control (Fernández et al., 2018) being preferred in the literature. Regardless, the intention with these manually-engineered spatial context features is to provide an approximation of spatial context, which is easy to implement, in order to contrast it with the generated spatial context embeddings in terms of classifier performance and behaviour. Given that such a measure has been successfully and usefully incorporated in previous literature, it was deemed appropriate for this present project.

4.4.2. *Spatial Context Embeddings*

4.5. Model Training

Decroos, Bransen, Van Haaren, & Davis (2019) use 4 different supervised classification techniques to estimate scoring and conceding probabilities, enabling VAEP calculations. This project will replicate the implementation of these same 4 techniques as closely as possible, albeit they do not provide comprehensive details on the Python version used, librarie versions, or all model hyperparameters.

Given that VAEP requires two concurrent classifiers to be trained for scoring and conceding probability estimates, respectively, and the introduction of manually engineered spatial features and spatial embeddings will result in a total of 32 individual classifier models being trained, enablining the construction of: baseline VAEP; baseline + manual features; baseline + embeddings; and baseline + manual features + embeddings models. This ablation design allows for the impact of adding spatial context on model performance and behaviour to be investigated.

4.5.1. *Logistic Regression*

Logistic regression (Cox, 1958) is a classification technique which estimates the probability that a given data instance belongs to some reference class (i.e. an action results in a goal scored/ conceded vs no goal scored/conceded, within k events). Logistic regression learns a vector of weights $\mathbf{w}$ and bias b which minimises cross-entropy loss and produces a probability estimate for the data instance beloning to the reference class:

$$P(Y = 1 \mid X) = \frac{1}{1 + e^{-(\mathbf{w}^\top \mathbf{x} + b)}} \tag{8}$$

Decroos, Bransen, Van Haaren, & Davis (2019) report the use of L2 Regularisation and Limited-memory BFGS, which is replicated in this project.

L2 regularisation introduces an additional penalty term to the loss function (in this case cross-entropy loss, see Section 4.5.4). This penality term is some multiple $\lambda \in \mathbb{R}^+$ of the sum of squared weights:

$$\text{L2} = \lambda \sum_{i=1}^{N} w_i^2 \tag{9}$$

`scikit-learn` uses a hyperparameter $C = \frac{1}{\lambda}$ to control regularisation strength, which is set to 1 by default. Since the authors do not specify otherwise, this default will be maintained for training.

Limited-memory BFGS (Liu & Nocedal, 1989) is a computationally-efficient optimisation method, used in logistic regression to iteratively minimise the loss function during training. Again, this hyperparameter is the default for `scikit-learn`.

4.5.2. *Random Forest*

Classification tasks may also be performed by decision trees. We can treat a classification task as a form of recursive partitioning of the dataset until we obtain maximally *pure* partitions primarily consisting of data instances from a single classification class. Choosing a feature and subsequent splitting threshold which minimises impurity (see Section 4.5.5) until a maximum tree depth or minimum leaf node size is reached builds a decision tree which can be used to make classifications by applying a sequence of logical filters.

A random forest (Breiman, 2001) is an ensemble method which aggregates results from multiple decision trees, each trained on a random subset of the training set. In a classification task, this aggregation is employed by majority voting. For each data instance, the class predictions made by each tree in the forest are counted up and the class with the most "votes" is outputted as the predicted class for that instance. By having class predictions be made via majority voting from multiple decision trees, a random forest benefits from being less prone to overfitting and more robust to outliers in the dataset.

Decroos, Bransen, Van Haaren, & Davis (2019) indicate that they used 100 trees and 40 parallel threads to train their random forest models. Thus this project also used 100 trees. However, the number of parallel threads simply determines the number of concurrent processes used, which does not impact model behaviour other than via its runtime. Since my hardware is different to the authors', this hyperparameter will be chosen during training and reported in the write up.

4.5.3. *Gradient Boosting*

Gradient boosting (Friedman, 2001) is another ensemble method involving decision trees. Gradient boosting methods construct a "strong" predictor by sequentially adding "weak" learners, where each new learner is trained to correct the errors of the previous ensemble.

$$\hat{y}_i = \sum_{t=1}^{T} \eta f_t(x_i), \quad f_t \in \mathcal{F} \tag{10}$$

Each function f_t is learned sequentially to minimise a loss function (like cross-entropy loss). At each iteration, the new weak learner is fit to the negative gradient of the loss with respect to the current model predictions. The learning rate η is used to determine how much the next tree contributes to model estimates.

4.5.3.1. *XGBoost*

XGBoost (Chen & Guestrin, 2016) builds upon standard gradient boosting by incorporating regularisation and introducing improved computational efficiency, even being capable of handling sparse data (i.e. data instances which are empty in some subset of features). Decroos, Bransen, Van Haaren, & Davis (2019) use 100 trees which are allowed to reach a maximum depth of 3 nodes. The learning rate η is equal to 0.1. Again, 40 parallel threads are used. Each of these hyperparameters (except the parallel threads) will be held constant in this project.

4.5.3.2. *CatBoost*

CatBoost (Prokhorenkova et al., 2019) is a gradient boosting method which is optimised for processing categorical predictors without requiring manual one-hot encoding. Decroos, Bransen, Van Haaren, & Davis (2019) retain the default hyperparameters for CatBoost (CatBoost Developers, n.d.), and again use 40 parallel threads.

4.5.4. *Loss Function: Cross-Entropy Loss*

Logistic regression, XGBoost and CatBoost each minimise cross-entropy loss in their classification predictions, according to their default implementations in the aforementioned Python libraries. Cross-entropy measures how "surprised" the model is by observations drawn from the true data-generating distribution $p(x)$, when evaluated under the predicted distribution $q(x)$. If the model tends to assign low probability to data instances which are probable under $p(x)$, then cross-entropy will be high. A low cross-entropy is found when $q(x)$ approximates $p(x)$ well.

$$H(p, q) = -\sum_{x \in X} p(x) \log q(x) \tag{11}$$

4.5.5. *Split Criterion: Gini impurity*

Whilst the split criterion used for the random forest models is not specified, the authors do indicate that they "used the implementation from the `scikit-learn` Python package" (Decroos, Bransen, Van Haaren, & Davis, 2019, p. 1860). It is thus inferred that they use the default split criterion of Gini impurity (Pedregosa et al., 2011; scikit-learn Developers, n.d.).

Gini impurity indicates the probability of assigning an incorrect label to a randomly chosen data instance by simply using the relative frequence of each label's occurrence in the data.

$$G_m = 1 - \sum_{k=1}^{K} \left(\frac{n_{m_k}}{n_m} \right)^2 \tag{12}$$

Decision trees within random forest classifiers, at each splitting decision, choose the feature and value threshold s which minimises the sum of the weighted Gini impurities of each partition m:

$$G_{m_{<s}/m_{>s}} = \frac{n_{m_{<s}}}{n_m} G_{m_{<s}} + \frac{n_{m_{>s}}}{n_m} G_{m_{>s}} \tag{13}$$

This results in the *purest* possible split and thus corresponds to the most discriminative splitting decision available. This process then iterates on each node in the decision tree until leaf nodes are reached which each only contain a single class (in this case, 0 or 1). Data instances can then be classified according to this derived decision tree.

4.6. Model Evaluation Metrics

4.6.1. *Area Under the Receiver Operating Characteristic Curve (AUC-ROC)*

A receiver operating characteristic curve (ROC) is produced by plotting the true-positive rate (TPR) against the false positive rate (FPR) for each probability threshold a model could use to classify a data instance into the reference class (i.e. the action results in a goal scored/conceded within k events).

4.6.2. *Brier Score*

Decroos, Bransen, Van Haaren, & Davis (2019) use the Brier score (Brier, 1950) to measure the performance of the VAEP classifier models by comparing the model's predicted probability of a data instance belonging to a class with its true class membership.

$$\mathrm{BS} = \frac{1}{N}\sum_{i=1}^{N}\sum_{k=1}^{K}\left(p_{ik} - y_{ik}\right)^2 \tag{14}$$

This is effectively equivalent to the mean squared error, where predictions equate to probabilities of belonging to the kth class and the true values are a one-hot encoding of that data instance's membership of that k^{th} class. For binary classification this simplifies to:

$$\mathrm{BS}_{\mathrm{binary}} = \frac{1}{N}\sum_{i=1}^{N}\left(p_i - y_i\right)^2 \tag{15}$$

A lower Brier score indicates that the model's predicted probabilities are, on average, closer to the true outcomes.

4.6.3. *Cross-Entropy Loss*

Cross-entropy loss was introduced as a loss function in Section 4.5.4, but it may also be used as a classification model evaluation metric. This project intends to use cross-entropy loss to evaluate each of the two classifier models which constitute each VAEP model.

4.6.4. *Expected Calibration Error (ECE)*

Expected calibration error (ECE) (Naeini et al., 2015) is a performance metric used to evaluate the ability of a classification model to make accurate predictions. ECE creates m bins between 0 and 1 and assigns predictions to bins based on their probability estimates. For each bin, we take the accuracy of predictions in that bin o_i and subtract from it the average of the probability estimates in that bin e_i. This is weighted by the proportion of predictions in that bin. Each bin's score is the summed to provide the ECE of the model's predictions. A lower ECE is more desirable, indicating that model predictions are close to true frequencies.

$$\mathrm{ECE} = \sum_{i=1}^{K} P(i) \cdot |o_i - e_i|$$

4.7. Model Interpretability

4.7.1. *Permutation Importance*

Permutation importance is a method for estimating feature importance by randomly permuting one or more features and evaluating the change in model performance metrics. Features which, when permuted, result in a larger drop in model performance can be inferred to be more important in model estimations. Care should be taken with multicolinearity, as permutations will not entirely disconnect their relationship with others model features since those relationships can continue to exist via their colinear counterparts.

Permutation importance will thus be used in this project to highlight which features or set of features are most determinative of the VAEP values assigned to actions. The potential prescence of multicolinarity could be address via hierarchical clustering

4.7.2. *Partial Dependence Plots (PDPs)*

Partial dependence plots (PDPs) (Friedman, 2001) are used to visualise the relationship between some set of features (usually just 1 or 2) and the predictions made by a machine learning model, holding other features constant. In this case, we can use PDP plots to explore our interpretable feature set, that is the baseline and manually-engineered features, as opposed to the spatial embeddings whose dimensions are not directly interpretable.

4.7.3. *k-Means Clustering*

We can also explore the spatial embeddings via k-means clustering, similarly to how DeepCluster applies k-means to image embeddings generated by a convolutional neural network (CNN) (Caron et al., 2019). k-means is an unsupervised clustering algorithm which finds optimal locations for some k centroids within the d-dimensional feature space. The algorithm initialises k centroids within the feature space, assigns data points to their nearest centroid and then recalculates the location of each centroid to the position which minimises the distance to its assigned data points. The data points are then reassigned and the algorithm iterates until convergence. A number of distance measures may be used, with euclidean distance being most common.

$$\text{dist } (x_i, x_n) = \sqrt{\sum_{d=1}^{D} \left(x_{i_d} - x_{n_d} \right)^2} \tag{16}$$

The benefit of using k-means clustering with the present project's spatial embeddings is that underlying relationships represented by these embeddings can be explored by comparing the properties of clusters. For instance, player position heatmaps for each cluster could be plotted to potentially reveal notable team shapes, or if certain clusters tend to involve certain action types over others.

4.7.3.1. *Choosing* k

By fitting a model to the data with varying values of k, we can use a number of metrics to evaluate how appropriate the chosen value of k is for clustering the given data points.

4.7.3.1.1. *Inertia*

The hyperparameter k can be chosen by considering the marginal reduction in *inertia* as k increases. A suitable k can be chosen at the point where this marginal reduction becomes negligible (this is known as the *elbow method*):

$$\text{Inertia} = \sum_{j=1}^{K} \sum_{x_i \in C_j} \parallel x_i - \mu_j \parallel^2 \tag{17}$$

Whilst commonly used in deriving k and a somewhat informative heuristic, the elbow method is inherently statistically arbitrary as a means for determining the appropriate number of clusters (Schubert, 2023).

4.7.3.1.2. *Average Silhouette Score*

An additional metric is the average *silhouette score* (Rousseeuw, 1987) for each value of k. For each data instance i, we can determine its average distance to others instances in the same cluster (known as *cohesion* $a(i)$) and its average distance to instances in the next nearest cluster (known as *separation* $b(i)$). Taking the differences between the separation and cohesion and dividing by the maximum of the two, we get the data instance's silhouette score.

If separation is lower than cohesion for a data instance, the silhouette score will be negative and thus suggests the data instance is closer to some other cluster(s) relative to its distance to its assigned cluster. Likewise, greater cohesion over separation suggests the data instance is appropriately assigned to its cluster, in terms of the distance metric used.

An average can then be taken across all data points and this average score can be compared across different values for k, where 1 indicates good overall clustering and -1 poor:

$$\text{SC} = \frac{1}{N}\sum_{i=1}^{N}\frac{b(i)-a(i)}{\max\{a(i),b(i)\}} \tag{18}$$

4.7.3.1.3. *Calinski-Harabasz Index (CHI)*

We may also appeal to the Calinski-Harabasz index (CHI) (Caliński & Harabasz, 1974) in assessing a suitable k. CHI is just a ratio of the between-cluster sum of squares (BCSS) and the within-cluster sum of squares (WCSS, i.e. the inertia), where the former is the sum of squared distances between cluster centroids and the latter is the sum of squared distances of each data point to its respective cluster. Each of these sum of squares measures are normalised by their respective degrees of freedom.

$$\text{CHI} = \frac{\frac{1}{k-1}\sum_{i=1}^{k} n_i \parallel c_i - c \parallel^2}{\frac{1}{n-k}\sum_{i=1}^{k}\sum_{x\in C_i}\parallel x - c_i \parallel^2} \tag{19}$$

4.7.3.1.4. *Davies-Bouldin Index (DBI)*

Lastly, the Davies-Bouldin index (DBI) provides a simultaneous evaluation of how separated each cluster is and how compact data instances are to their assigned cluster. For each pair of clusters i, j, we can consider how their individual compactness ($S_i and S_j$) and their relative distances compare by taking the ratio of the sum of their compactness to the distance of their centroids.

$$R_{ij} = \frac{S_i + S_j}{M_{ij}} \tag{20}$$

S_i and M_{ij} can, in principle, be any measure of internal compactness or cluster separation, respectively. `scikit-learn` implements the average Euclidean distance between data instances and their cluster's centroid for internal compactness, and Euclidean distance between cluster centroids for cluster separation.

A lower value for R_{ij} indicates the clusters i, j are well-separated and internally compact. As such, for each cluster i, we can consider its *worst-case scenario* by finding its corresponding j^{th} cluster which maximises R_{ij}. Taking the sum of each of these maximum R_{ij} values and dividing

by the number of clusters k gives a measure of the average clustering performance $\bar{R}$, where minimising the measure optimises clustering performance:

$$\bar{R} = \frac{1}{k} \sum_{i=1}^{k} \max_{i \neq j} R_{ij} \tag{21}$$

4.7.4. *Uniform Manifold Approximation and Projection (UMAP)*

In order to interpret the spatial embeddings, uniform manifold approximation and projection (UMAP) (McInnes et al., 2020) can be used as a dimensionality reduction technique to represent embeddings in a 2 or 3 dimensional space, which can then be visualised. The color of data points within a UMAP can then be determined by properties such as the VAEP value, scoring/ conceding probability or k-means clusters.

5. Results

5.1. Set Transformer Training

trial	value	batch size	output dim	flag weight	hidden dim	drop -out	lr	num heads	num layers	weight decay	xy weight
0	0.220	1024	64	0.375	32	0.1	2.8e-03	2	1	1.5e-04	1.42
1	0.140	512	16	0.326	64	0.1	6.6e-04	8	2	9.1e-04	1.28
2	0.112	512	64	0.057	64	0.0	3.2e-03	8	2	7.6e-05	1.59
3	0.128	256	16	0.205	32	0.2	1.8e-04	2	3	1.6e-05	0.86
4	0.244	512	64	0.456	64	0.2	2.3e-04	8	2	7.5e-06	1.28
5	0.280	1024	32	0.428	32	0.3	1.8e-03	8	3	1.4e-06	1.56
6	0.303	1024	16	0.459	64	0.3	1.4e-04	2	1	3.8e-04	1.73
7	0.083	256	32	0.105	256	0.2	1.1e-04	8	2	1.7e-04	1.21
8	0.791	256	16	0.463	256	0.1	4.8e-03	8	4	5.1e-06	1.88
9	1.683	1024	32	0.409	128	0.3	4.3e-03	4	3	1.1e-05	1.70
10	0.053	256	32	0.055	256	0.2	5.0e-04	4	4	5.8e-05	0.57
11	0.046	256	32	0.057	256	0.2	4.6e-04	4	4	5.7e-05	0.55
12	0.079	256	32	0.130	256	0.2	4.7e-04	4	4	3.6e-05	0.56
13	0.091	256	32	0.191	256	0.2	7.7e-04	4	4	3.3e-05	0.53
14	0.060	256	32	0.054	256	0.0	3.5e-04	4	4	5.3e-05	0.87
15	0.101	256	32	0.158	128	0.2	1.2e-03	4	4	1.5e-04	0.78
16	0.096	256	32	0.238	256	0.2	3.2e-04	4	4	2.4e-05	0.72
17	0.118	256	32	0.097	256	0.2	1.0e-03	4	4	3.0e-06	1.03
18	0.057	256	32	0.051	256	0.0	5.1e-04	4	4	7.5e-05	0.65
19	0.158	512	64	0.277	256	0.2	2.5e-04	4	1	3.7e-04	1.03
20	0.109	256	32	0.145	128	0.2	1.2e-03	4	4	1.8e-05	1.00
21	0.213	256	32	0.053	256	0.0	5.2e-04	4	4	8.1e-05	0.67
22	0.051	256	32	0.095	256	0.0	4.4e-04	4	4	6.7e-05	0.51
23	0.055	256	32	0.105	256	0.0	3.6e-04	4	4	3.1e-04	0.51
24	0.076	256	32	0.093	256	0.0	7.0e-04	4	4	4.7e-05	0.62
25	0.093	256	32	0.169	256	0.0	4.4e-04	2	4	1.1e-04	0.80
26	0.063	256	32	0.122	256	0.1	2.6e-04	4	4	2.4e-04	0.70

trial	value	batch size	output dim	flag weight	hidden dim	drop -out	lr	num heads	num layers	weight decay	xy weight
27	0.094	256	32	0.075	128	0.3	1.7e-04	4	3	2.2e-05	0.92
28	0.109	512	16	0.201	32	0.2	8.4e-04	4	1	5.5e-05	0.50
29	0.100	1024	64	0.145	32	0.1	1.7e-03	2	1	7.5e-04	0.61

Table 1: Optuna Hyperparameter Tuning Results

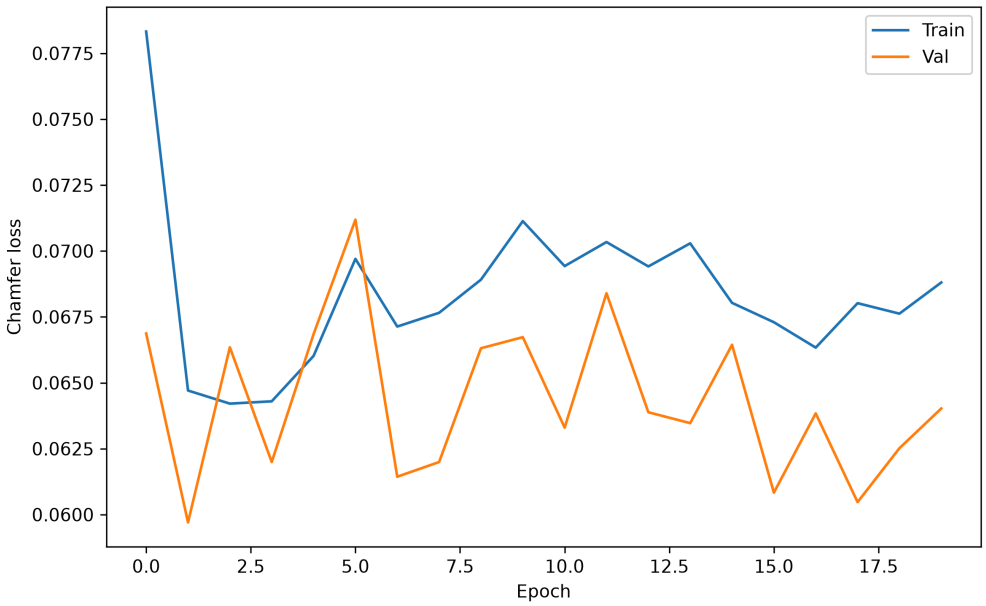


Figure 3: Set Transformer Training Loss Curve

5.2. Model Training

model	feature set	target	brier score	log loss	ece	auc
CatBoost	all	concedes	0.001125	0.007327	0.000477	0.887021
CatBoost	baseline	concedes	0.001167	0.007428	0.000423	0.890564
CatBoost	embeddings	concedes	0.001142	0.007412	0.000451	0.890103
CatBoost	manual	concedes	0.001173	0.007635	0.000473	0.883410
LR	all	concedes	0.001175	0.007585	0.000181	0.885107
LR	baseline	concedes	0.001192	0.007853	0.000101	0.872270
LR	embeddings	concedes	0.001177	0.007640	0.000088	0.881908
LR	manual	concedes	0.001188	0.007694	0.000141	0.877531
RF	all	concedes	0.001239	0.008040	0.000227	0.894979
RF	baseline	concedes	0.001240	0.008171	0.000241	0.887575
RF	embeddings	concedes	0.001228	0.008096	0.000192	0.882123
RF	manual	concedes	0.001234	0.008056	0.000272	0.894220
XGBoost	all	concedes	0.001147	0.006937	0.000145	0.913801
XGBoost	baseline	concedes	0.001179	0.007199	0.000120	0.897184
XGBoost	embeddings	concedes	0.001143	0.006871	0.000110	0.917760
XGBoost	manual	concedes	0.001156	0.006968	0.000196	0.906989

Table 2: Concedes Classifiers Test Results

model	feature set	target	brier score	log loss	ece	auc
CatBoost	all	scores	0.005960	0.032472	0.001515	0.838349
CatBoost	baseline	scores	0.005996	0.033541	0.001110	0.807787
CatBoost	embeddings	scores	0.005960	0.032670	0.001455	0.834615
CatBoost	manual	scores	0.006009	0.033148	0.001314	0.824931
LR	all	scores	0.005905	0.031800	0.000799	0.853255
LR	baseline	scores	0.005955	0.033980	0.000781	0.772612
LR	embeddings	scores	0.005900	0.031748	0.000936	0.854696
LR	manual	scores	0.005942	0.032962	0.000694	0.816833
RF	all	scores	0.006463	0.035154	0.002268	0.805724
RF	baseline	scores	0.006472	0.037071	0.002074	0.755487
RF	embeddings	scores	0.006380	0.035205	0.002037	0.799344
RF	manual	scores	0.006514	0.036883	0.002111	0.756424
XGBoost	all	scores	0.005869	0.031383	0.000787	0.856037
XGBoost	baseline	scores	0.005881	0.032361	0.000785	0.821213
XGBoost	embeddings	scores	0.005857	0.031320	0.000849	0.855174
XGBoost	manual	scores	0.005879	0.031918	0.000718	0.839078

Table 3: Scores Classifiers Test Results

5.2.1. *Feature Set Significance Tests*

5.2.1.1. *Concedes Classifier*

comparison	z	p holm	delta auc	auc 1	auc 2
all vs baseline	0.53	1	−3.54e-03	0.8870	0.8906
all vs embeddings	0.63	1	−3.08e-03	0.8870	0.8901
all vs manual	−0.65	1	3.61e-03	0.8870	0.8834
baseline vs embeddings	−0.08	1	4.61e-04	0.8906	0.8901
baseline vs manual	−1.25	1	7.15e-03	0.8906	0.8834
embeddings vs manual	−1.07	1	6.69e-03	0.8901	0.8834

Table 4: Concedes Model Feature Set Significance Test Comparison (CatBoost)

comparison	z	p holm	delta auc	auc 1	auc 2
all vs baseline	−1.57	0.7	1.28e-02	0.8851	0.8723
all vs embeddings	−1.34	0.9	3.20e-03	0.8851	0.8819
all vs manual	−1.25	0.9	7.58e-03	0.8851	0.8775
baseline vs embeddings	1.14	0.9	−9.64e-03	0.8723	0.8819
baseline vs manual	1.04	0.9	−5.26e-03	0.8723	0.8775
embeddings vs manual	−0.67	0.9	4.38e-03	0.8819	0.8775

Table 5: Concedes Model Feature Set Significance Test Comparison (Logistic Regression)

comparison	z	p holm	delta auc	auc 1	auc 2
all vs baseline	−0.93	1	7.40e-03	0.8950	0.8876
all vs embeddings	−3.25	0.007	1.29e-02	0.8950	0.8821
all vs manual	−0.14	1	7.59e-04	0.8950	0.8942
baseline vs embeddings	−0.64	1	5.45e-03	0.8876	0.8821
baseline vs manual	0.97	1	−6.64e-03	0.8876	0.8942
embeddings vs manual	1.75	0.4	−1.21e-02	0.8821	0.8942

Table 6: Concedes Model Feature Set Significance Test Comparison (Random Forest)

comparison	z	p holm	delta auc	auc 1	auc 2
all vs baseline	−2.90	0.02	1.66e-02	0.9138	0.8972
all vs embeddings	1.76	0.2	−3.96e-03	0.9138	0.9178
all vs manual	−1.37	0.2	6.81e-03	0.9138	0.9070
baseline vs embeddings	3.75	0.001	−2.06e-02	0.8972	0.9178
baseline vs manual	2.18	0.1	−9.81e-03	0.8972	0.9070
embeddings vs manual	−2.20	0.1	1.08e-02	0.9178	0.9070

Table 7: Concedes Model Feature Set Significance Test Comparison (XGBoost)

5.2.1.2. *Scores Classifier*

comparison	z	p holm	delta auc	auc 1	auc 2
all vs baseline	−6.31	2e-09	3.06e-02	0.8383	0.8078
all vs embeddings	−1.17	0.2	3.73e-03	0.8383	0.8346
all vs manual	−3.16	0.005	1.34e-02	0.8383	0.8249
baseline vs embeddings	5.84	3e-08	−2.68e-02	0.8078	0.8346
baseline vs manual	4.40	4e-05	−1.71e-02	0.8078	0.8249
embeddings vs manual	−2.19	0.06	9.68e-03	0.8346	0.8249

Table 8: Scores Model Feature Set Significance Test Comparison (CatBoost)

comparison	z	p holm	delta auc	auc 1	auc 2
all vs baseline	−13.42	2e-40	8.06e-02	0.8533	0.7726
all vs embeddings	1.13	0.3	−1.44e-03	0.8533	0.8547
all vs manual	−10.87	6e-27	3.64e-02	0.8533	0.8168
baseline vs embeddings	13.97	1e-43	−8.21e-02	0.7726	0.8547
baseline vs manual	8.41	8e-17	−4.42e-02	0.7726	0.8168
embeddings vs manual	−10.03	3e-23	3.79e-02	0.8547	0.8168

Table 9: Scores Model Feature Set Significance Test Comparison (Logistic Regression)

comparison	z	p holm	delta auc	auc 1	auc 2
all vs baseline	−6.58	3e-10	5.02e-02	0.8057	0.7555
all vs embeddings	−3.88	0.0002	6.38e-03	0.8057	0.7993
all vs manual	−6.33	1e-09	4.93e-02	0.8057	0.7564
baseline vs embeddings	5.74	4e-08	−4.39e-02	0.7555	0.7993
baseline vs manual	0.19	0.9	−9.36e-04	0.7555	0.7564
embeddings vs manual	−5.27	4e-07	4.29e-02	0.7993	0.7564

Table 10: Scores Model Feature Set Significance Test Comparison (Random Forest)

comparison	z	p holm	delta auc	auc 1	auc 2
all vs baseline	−8.80	7e-18	3.48e-02	0.8560	0.8212
all vs embeddings	−0.69	0.5	8.63e-04	0.8560	0.8552
all vs manual	−5.26	4e-07	1.70e-02	0.8560	0.8391
baseline vs embeddings	8.84	6e-18	−3.40e-02	0.8212	0.8552
baseline vs manual	6.03	7e-09	−1.79e-02	0.8212	0.8391
embeddings vs manual	−4.73	5e-06	1.61e-02	0.8552	0.8391

Table 11: Scores Model Feature Set Significance Test Comparison (XGBoost)

5.3. Model Interpretation

5.3.1. *Permutation Importance*

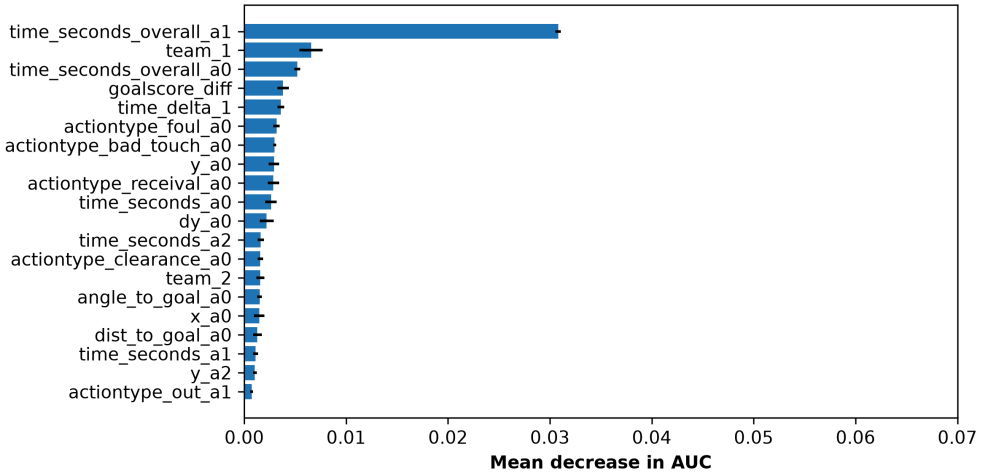


Figure 4: Concede Classifier Permutation Importance: Baseline Features

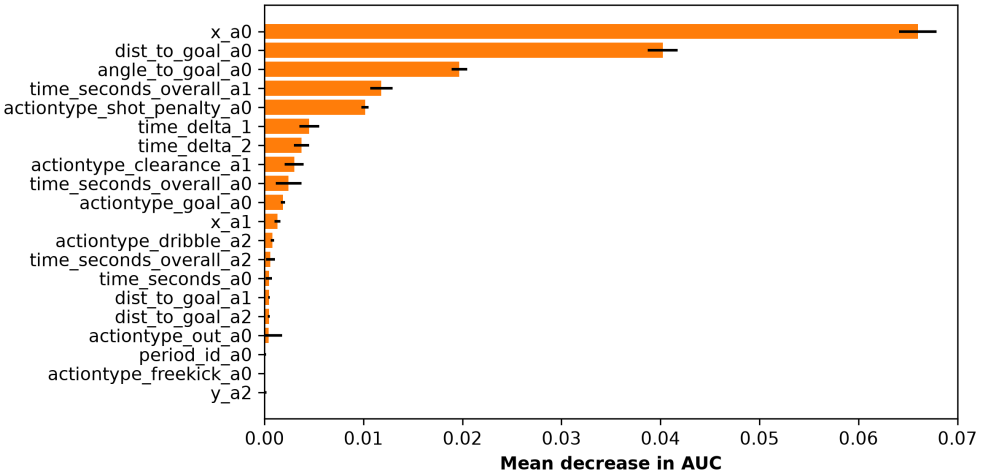


Figure 5: Score Classifier Permutation Importance: Baseline Features

5.3.2. *Partial Dependence Plots*

5.3.3. *K-means Clustering: Spatial Embeddings*

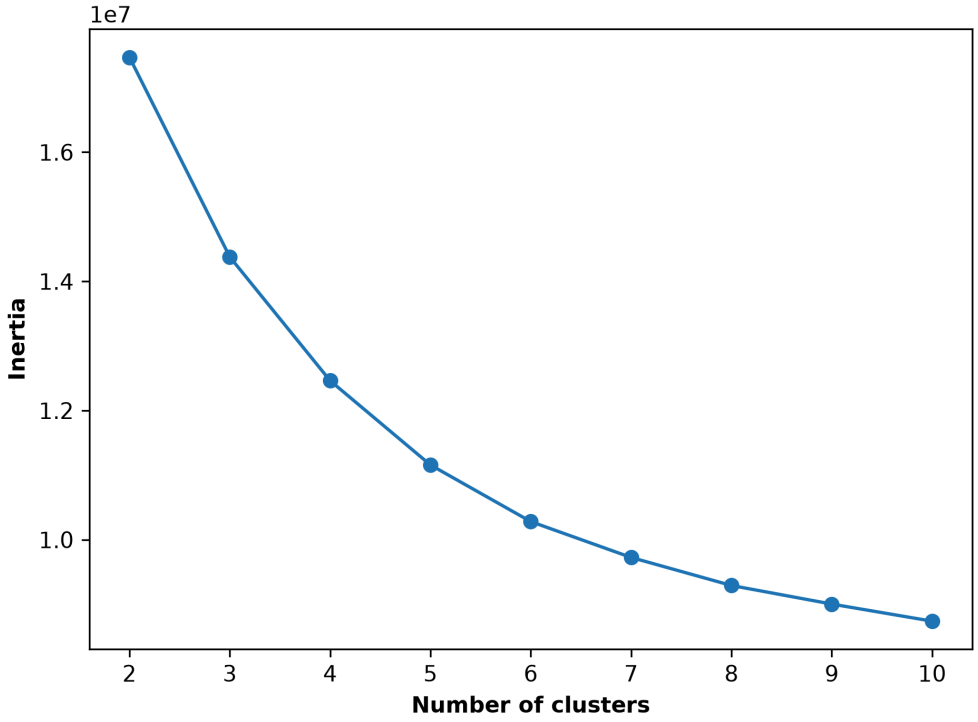


Figure 6: K-means Elbow Method Plot

k	inertia	silhouette	calinski harabasz	davies bouldin
2	1.75e+07	0.26	2.73e+05	1.60
3	1.44e+07	0.30	2.46e+05	1.35
4	1.25e+07	0.29	2.27e+05	1.49
5	1.12e+07	0.29	2.12e+05	1.47
6	1.03e+07	0.29	1.97e+05	1.47
7	9.72e+06	0.29	1.81e+05	1.59
8	9.29e+06	0.28	1.67e+05	1.69
9	9.00e+06	0.27	1.54e+05	1.74
10	8.74e+06	0.27	1.43e+05	1.77

Table 12: K-means Hyperparameter Results

5.3.4. *UMAP*

6. Discussion

7. Conclusion

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